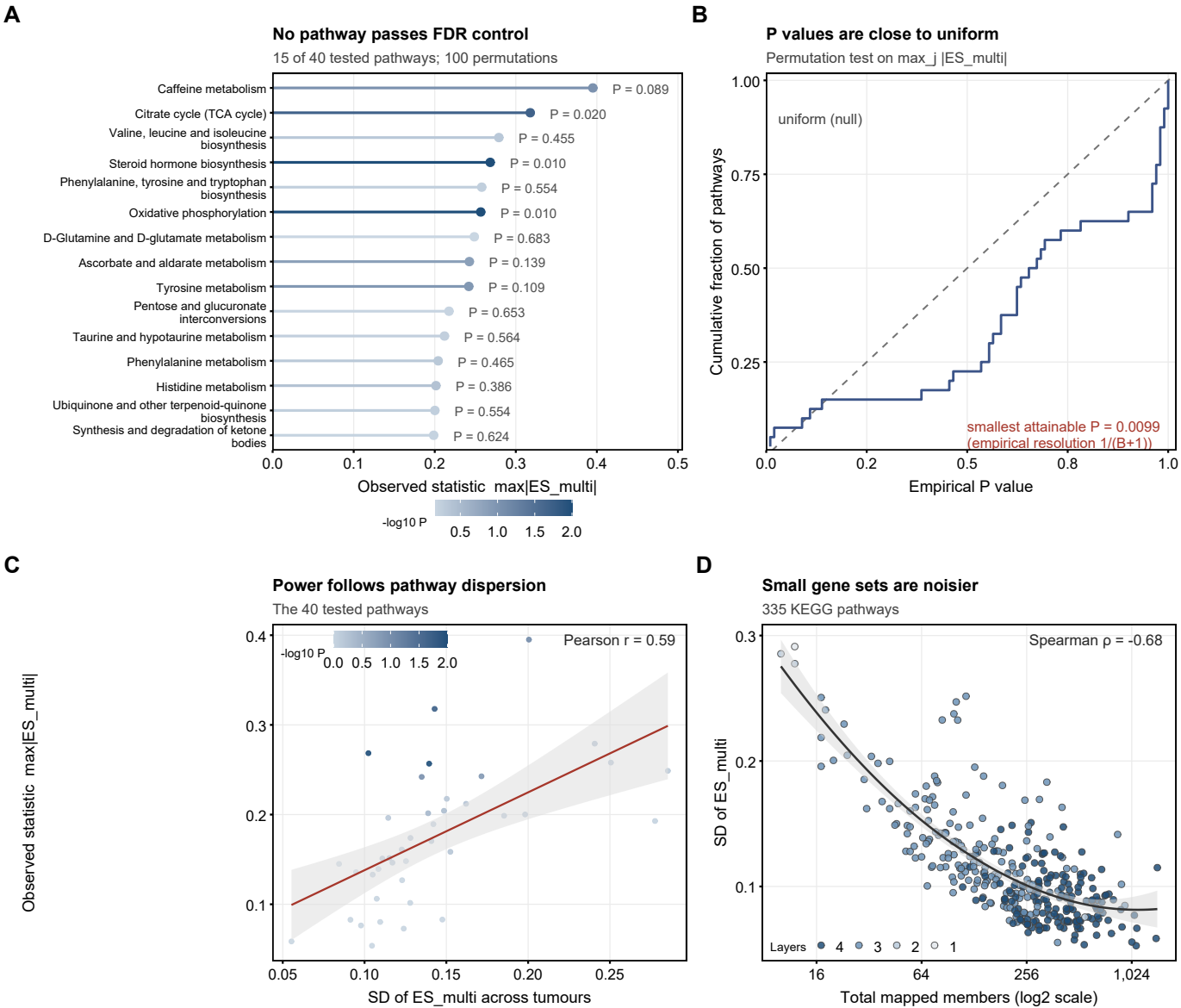

A
No pathway passes FDR control
15 of 40 tested pathways; 100 permutations
Caffeine metabolism
Citrate cycle (TCA cycle)
Valine, leucine and isoleucine biosynthesis
Steroid hormone biosynthesis
Phenylalanine, tyrosine and tryptophan biosynthesis
Oxidative phosphorylation
D-Glutamine and D-glutamate metabolism
Ascorbate and aldarate metabolism
Tyrosine metabolism
Pentose and glucuronate interconversions
Taurine and hypotaurine metabolism
Phenylalanine metabolism
Histidine metabolism
Ubiquinone and other terpenoid-quinone biosynthesis
Synthesis and degradation of ketone bodies
P = 0.089
P = 0.020
P = 0.455
P = 0.010
P = 0.554
P = 0.010
P = 0.683
P = 0.139
P = 0.109
P = 0.653
P = 0.564
P = 0.465
P = 0.386
P = 0.554
P = 0.624
0.0 0.1 0.2 0.3 0.4 0.5
Observed statistic max|ES_multi|
-log10 P
0.5 1.0 1.5 2.0
B
P values are close to uniform
Permutation test on max_j |ES_multi|
uniform (null)
smallest attainable P = 0.0099
(empirical resolution 1/(B+1))
Cumulative fraction of pathways
1.00 0.75 0.50 0.25
0.0 0.2 0.5 0.8 1.0
Empirical P value
C
Power follows pathway dispersion
The 40 tested pathways
-log10 P
0.0 0.5 1.0 1.5 2.0
Pearson r = 0.59
Observed statistic max|ES_multi|
0.4 0.3 0.2 0.1
0.05 0.10 0.15 0.20 0.25
SD of ES_multi across tumours
D
Small gene sets are noisier
335 KEGG pathways
Spearman ρ = -0.68
SD of ES_multi
0.3 0.2 0.1
Layers 4 3 2 1
16 64 256 1,024
Total mapped members (log2 scale)